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Questions

Q1.

* The photograph shows a toy car that contains a spring. Pulling the car backwards along the floor compresses the spring. Energy is stored in the compressed spring.



The car is released and the spring returns to its original state. There is a forward force on the car from the floor and the car moves forwards.

Explain why the floor exerts a forward force on the car and how this force affects the motion of the car as the spring returns to its original state.

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(Total for question = 6 marks)

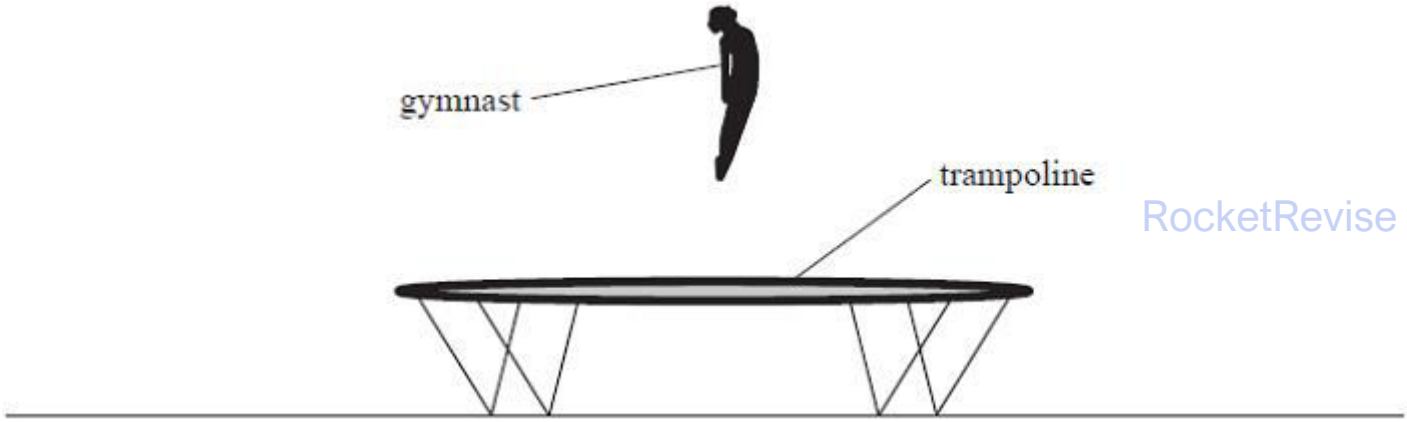
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Q2.

A gymnast bounces on a trampoline.

For part of each bounce, the gymnast is in contact with the trampoline. For the rest of each bounce the gymnast is in the air, as shown.



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(a) The trampoline gives the gymnast a maximum upward acceleration of 14.2 m s^{-2}.

Calculate the maximum upward force of the trampoline on the gymnast.
mass of gymnast = 58 kg

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Maximum upward force = ...

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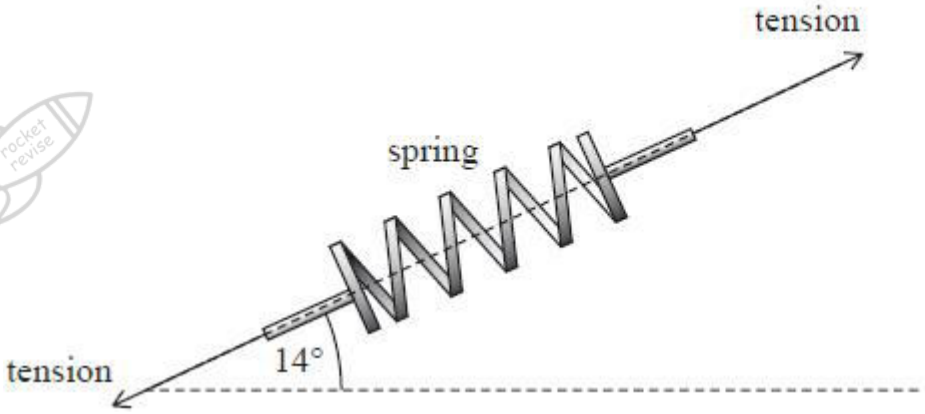
(b) The trampoline is made of a sheet of material attached to a frame by springs.

The vertical components of the tension in the springs provide the upward force on the gymnast.
The vertical component of the tension in one spring is 68 N when the spring makes an angle to the horizontal of 14°, as shown below.

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Not to scale



(i) Show that the tension in the spring is about 300 N.

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(ii) The extension of the spring was 4.6×10^{-2} m.
Calculate the stiffness of the spring.

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Stiffness = ...

*(c) The vertical acceleration of the gymnast varies while she is in contact with the trampoline.

Explain how the forces on the gymnast affect the vertical acceleration while she is in contact with the trampoline.
Your answer should identify the forces acting on the gymnast and the directions of the forces. Ignore air resistance.

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(Total for question = 14 marks)

Q3.

Which of the following is a vector quantity?

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- ☒ **A** density
- ☒ **B** kinetic energy
- ☒ **C** momentum
- ☒ **D** viscosity

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct (☒). If you change your mind, put a line through the box (☒) and then mark your new answer with a cross (☒).

When compressed by a force of 50 N, a spring had a length of 12.0 cm.

When compressed by a force of 70 N, the same spring had a length of 7.1 cm.

Which of the following expressions gives the stiffness k of the spring in N cm^{-1}?

- ☒ **A** $\frac{70}{12.0} - \frac{50}{7.1}$

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- ☒ **B** $\frac{(70 - 50)}{(12.0 - 7.1)}$
- ☒ **C** $\frac{50}{7.1} - \frac{70}{12.0}$
- ☒ **D** $\frac{(70 - 50)}{(12.0 - 4.9)}$

(Total for question = 1 mark)

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Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two identical springs are arranged in series. A force of 45.0 N is applied to the springs, as shown.



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The springs extend a total distance of 15 mm.

Which of the following expressions gives the stiffness of each spring in N mm^{-1}?

☐ **A** $\frac{15}{2 \times 45}$

☐ **B** $\frac{45}{2 \times 15}$

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☐ **C** $\frac{2 \times 15}{45}$

☐ **D** $\frac{2 \times 45}{15}$

(Total for question = 1 mark)

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Q6.

A student is investigating a material in the form of a wire.

Which of the following properties of the wire will change if a longer wire is used?

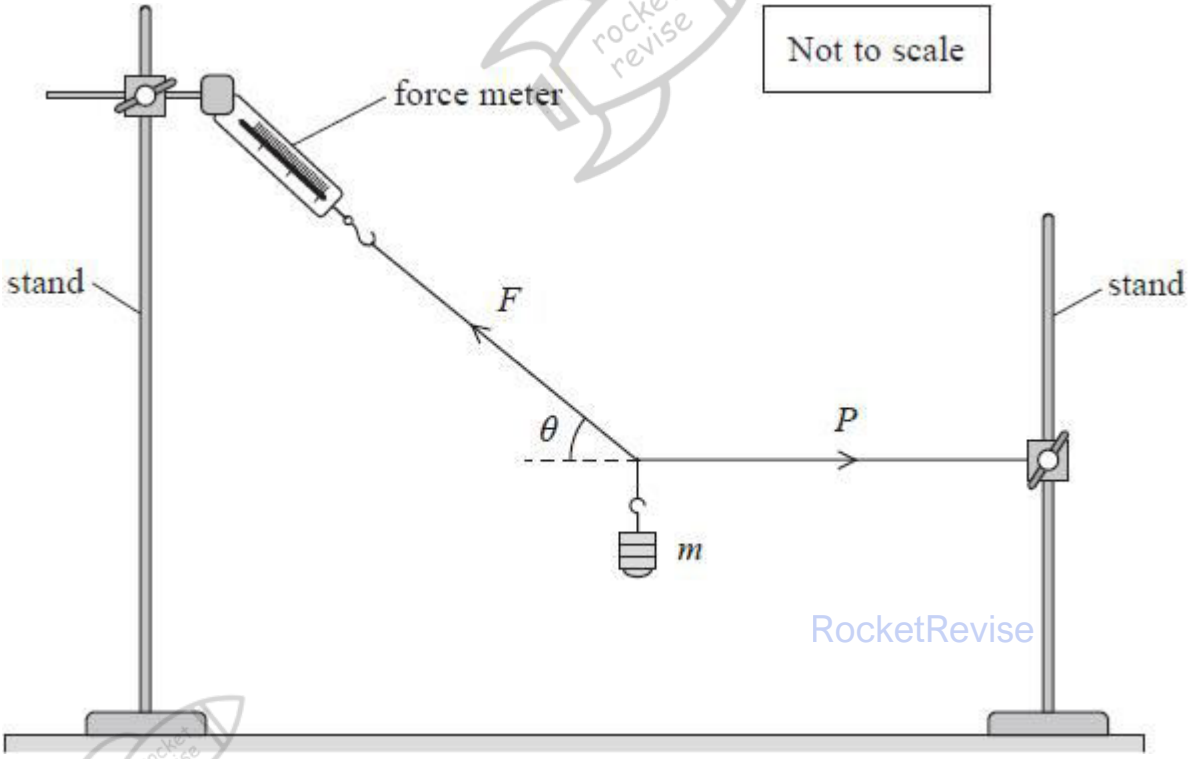
- ☒ **A** breaking stress
- ☒ **B** density
- ☒ **C** stiffness
- ☒ **D** Young modulus

(Total for question = 1 mark)

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Q7.

A mass *m* is held in equilibrium by strings attached to two clamp stands. The force meter records the force *F* in the upper string. The force in the horizontal string is *P*. The angle made by the upper string to the horizontal is *θ*, as shown.



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(a) The force meter allows force to be measured by means of Hooke's law.

The extension of the spring inside the force meter allows the stretching force to be read from a scale. When the force applied to stretch the spring is 15 N the extension of the spring is 8.0 cm. Show that the stiffness of the spring is about 2 N cm^{-1}.

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(b) When m is equal to 0.55 kg, the value of P is 8.5 N.

Calculate the value of θ, and the extension of the spring in the force meter.

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θ = ...

Extension of the spring = ...

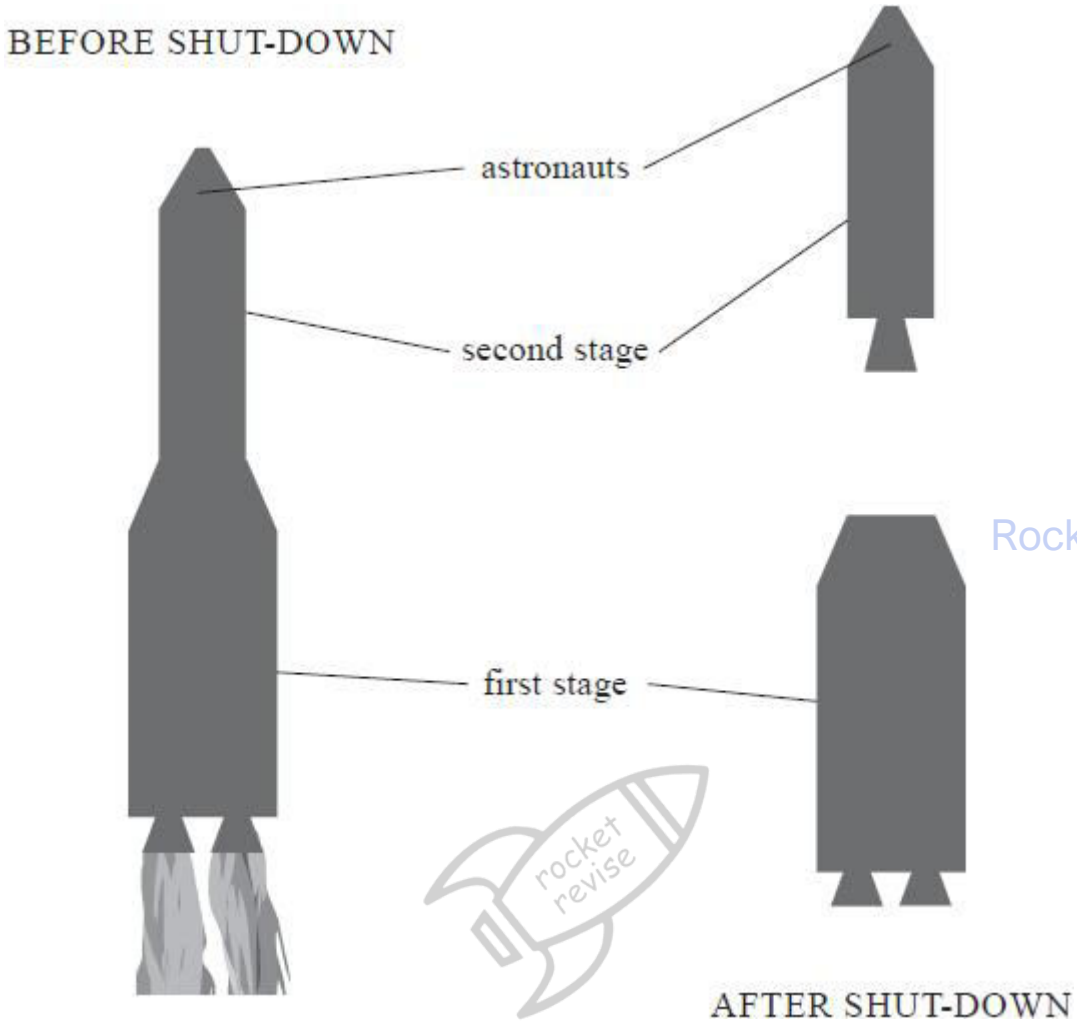
(Total for question = 8 marks)

Q8.

*A large spacecraft is made up of several 'stages'. Each stage consists of a rocket motor and a fuel tank. Once a stage has used all its fuel, the rocket motor in that stage shuts down. The stage then disconnects from the spacecraft and falls back to Earth.

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As the rocket is rising due to the upward force of the first stage, an astronaut feels himself pushed further and further back, compressing the back of his seat.

When the first stage shuts down, the astronaut is suddenly projected forwards by the seat. The astronaut is held in place by a safety strap.

Explain the effects experienced by the astronaut. You may assume that the force provided by the first stage rocket motor is constant until the moment it shuts down.

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Q9.

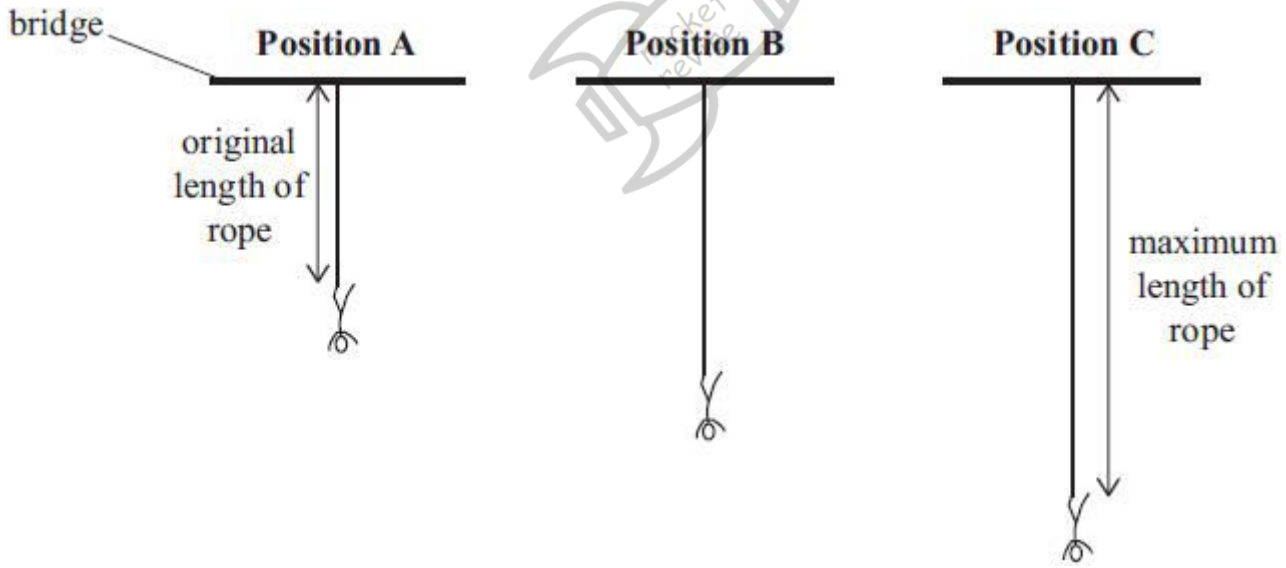
A student is about to do a bungee jump from a high bridge. The student is attached to the bridge by an elastic rope.

The student jumps from the bridge and initially accelerates. The diagrams show the student at three different positions during the bungee jump.

At **Position A** the distance fallen by the student is equal to the original length of the rope.

At **Position B** the student is in equilibrium.

At **Position C** the rope has maximum extension.



*(a) Explain how the acceleration of the student changes as she moves from **Position A** to **Position C**. Assume that air resistance is negligible.

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(b) The bridge is above a river. The student will reach the river if the rope stretches to a total length of 35 m during the bungee jump.

The student has a mass of 65 kg.
The original length of the rope is 17 m.
The rope behaves like a spring with stiffness constant 250 N m^{-1}.
Deduce whether the student will reach the river during the bungee jump.

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(Total for question = 10 marks)

Q10.

A student has a spring of length 0.18 m. The spring has a stiffness constant of 25 N m^{-1}.

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The student applies a force of 7.0 N to stretch the spring.

Calculate the new length of the spring.

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New length = ...

(Total for question = 2 marks)

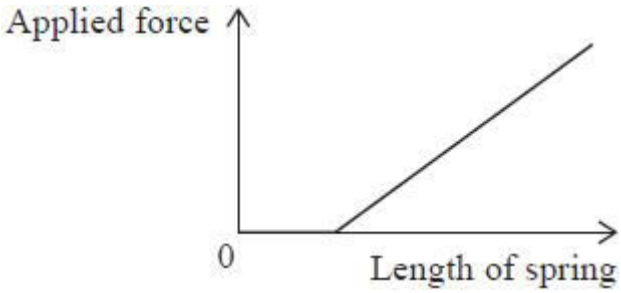
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Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A force is applied to stretch a spring. The graph shows how the length of the spring varies with the applied force.



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Which of the following gives the stiffness of the spring?

☐ **A** the area between the graph and the x-axis

☐ **B** the area between the graph and the y-axis

☐ **C** the gradient of the graph

☐ **D** $\dfrac{1}{\text{gradient of the graph}}$

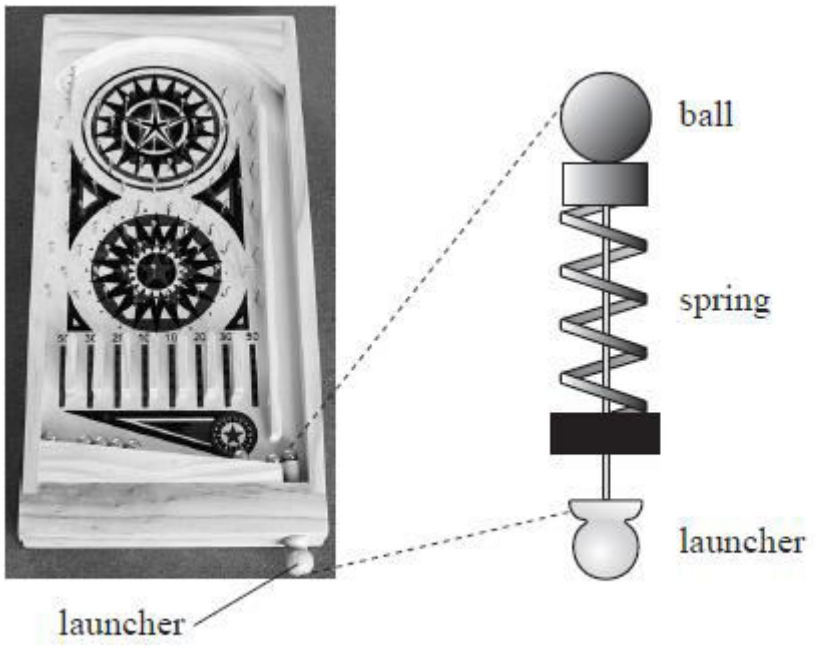
(Total for question = 1 mark)

Q12.

The photograph shows a toy pinball machine. The launcher is pulled back, compressing a spring. The spring obeys Hooke's law. When the launcher is released, the spring returns to its original length and a small ball is launched horizontally into the machine.

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(Source: © Valery Voennyy/Alamy Stock Photo)

When the launcher is pulled back, the spring is compressed by 5.0 cm. When the spring is released, the ball is launched at a speed of 8.0 cm s^{-1}.

(a) Show that the kinetic energy of the ball just after launching is about 4×10^{-5} J.

mass of ball = 12 g

(2)

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(b) Determine the force on the ball when the spring is released.

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Force = ...

(c) Determine the stiffness of the spring.

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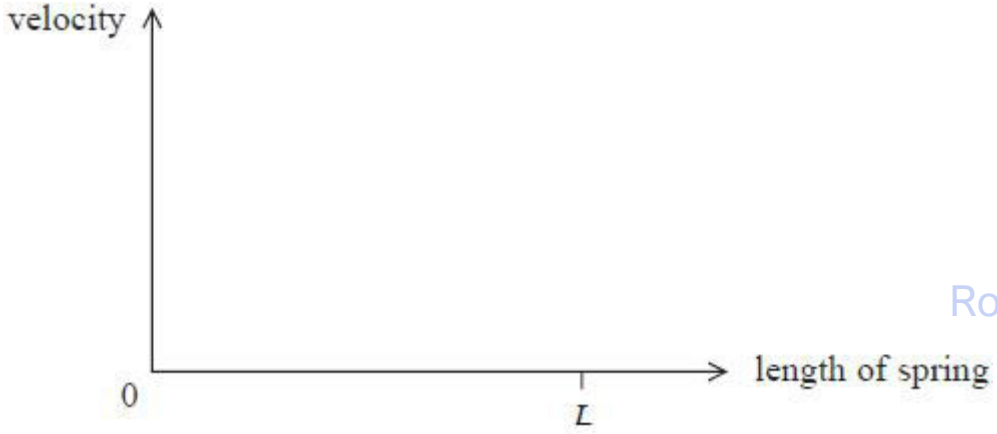
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Stiffness = ...

(d) The spring returns to its original length *L*.

Sketch a graph, on the axes below, to show how the velocity of the ball varies with the length of the spring.

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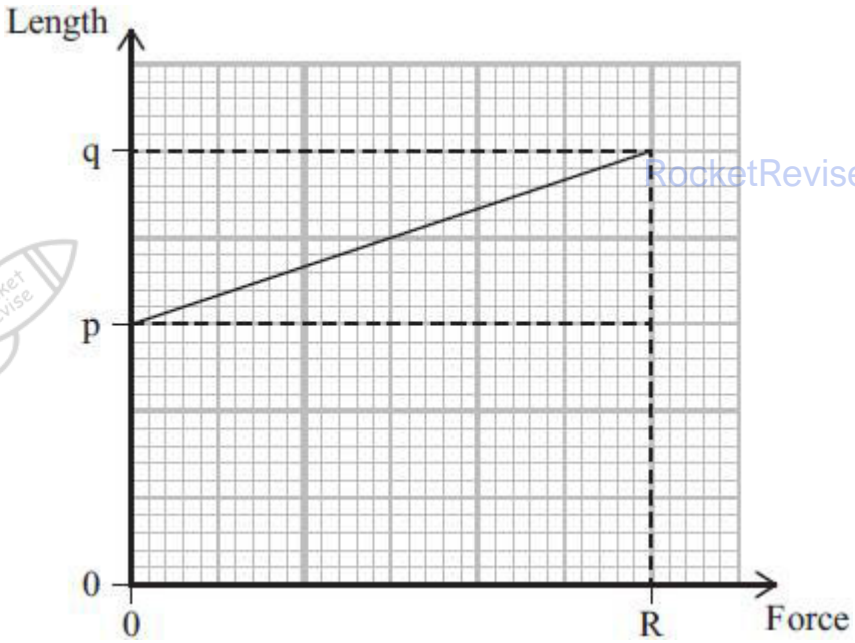
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(Total for question = 10 marks)

Q13.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student stretches a spring of original length p to a length q by applying an increasing force. The graph shows how the length of the spring changes as the student increases the force from zero to a magnitude R.



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Which of the following expressions gives the stiffness of the spring?

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☐ **A** $\frac{q - p}{R}$

☐ **B** $\frac{q}{R}$

☐ **C** $\frac{R}{q}$

☐ **D** $\frac{R}{q - p}$

(Total for question = 1 mark)

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Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When a force *F* is applied to a spring of stiffness *k*, the extension of the spring is Δx.

A force $2F$ is applied to a second spring of stiffness $3k$.

Which of the following expressions gives the extension of the second spring?

☐ **A** $\frac{2}{3}\Delta x$

☐ **B** $\frac{3}{2}\Delta x$

☐ **C** $\frac{1}{6}\Delta x$

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☐ **D** $6\Delta x$

(Total for question = 1 mark)

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